

Lab 00: Introduction

Objective

The purpose of this Lab. is to familiarize student with the programming environment they will be going to using throughout this course. This Lab. introduces the basics of NetBeans IDE and demonstrates how IDE software facilitates the process of programming. The second and most important purpose is for the students to use this IDE to solve four short programming exercises, which will allow the student to become comfortable with solving problems within the NetBeans environment.

Current Lab Learning Outcomes (LLO)

By completion of the lab the students should be able to know how to download, install and use NetBeans IDE.

1. Further they will be aware of how IDE software facilitates the process of programming.
2. The students will write their first java program and do further exercises.
3. Students will know the difference between "**System.out.print**" and "**System.out.println**".
4. Students will know introductory use of "**\n**" and "**\t**" within print statements

Lab Requirements

NetBeans IDE

Lab Description

Theory Review

The output statement has the following syntax:

System.out.println("Any Message to be printed");

- This statement will display the message written inside " " and moves the cursor to next line such that if there are another output statement(s) it will display the new message from the new line.

Another syntax for this statement is:

System.out.print("Any Message to be printed");

- This statement will display the message written inside " " and the cursor stays on the same line such that if there are another output statement(s) it will display the new message on the same line.

Installation Part 1:

Steps to Download Java and Netbeans

- 1- Open Internet explorer

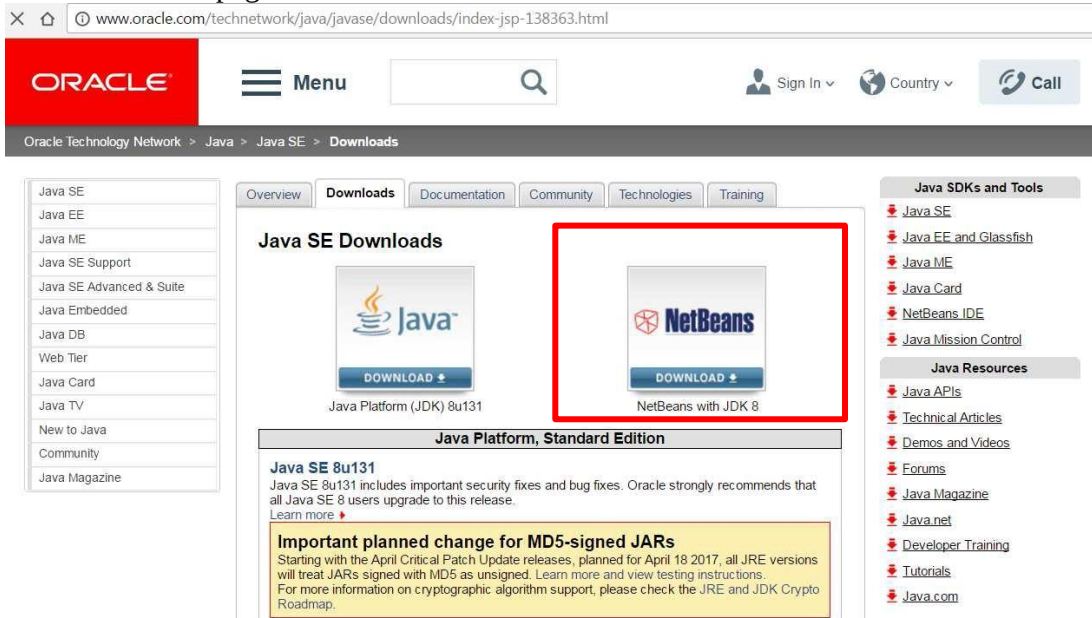


2- Go to google www.google.com and write **JDK download**

3- From the new page, click on the first result

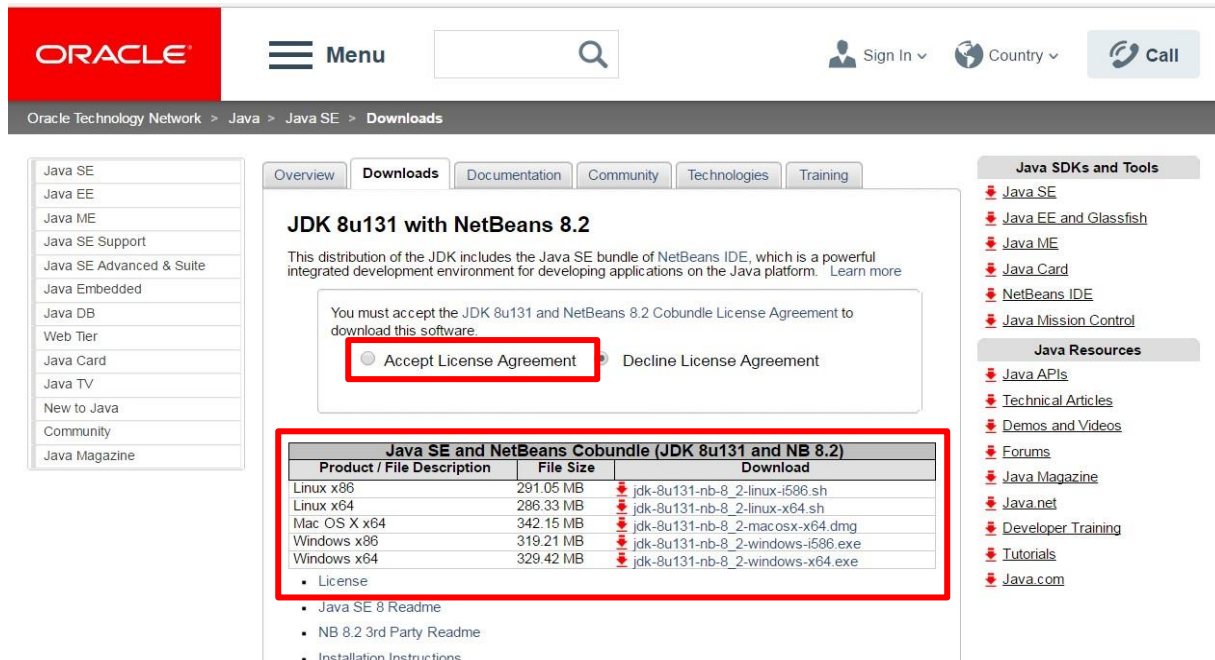


4- From the new page select "NetBeans with JDK"



5- From the next page choose the "Accept License Agreement" option and select the version suitable for your PC.

The file will start downloading. When downloading process is over, you can install the program following the steps given in next section.



JDK 8u131 with NetBeans 8.2

This distribution of the JDK includes the Java SE bundle of NetBeans IDE, which is a powerful integrated development environment for developing applications on the Java platform. [Learn more](#)

You must accept the JDK 8u131 and NetBeans 8.2 Cobundle License Agreement to download this software.

☒ Accept License Agreement ☐ Decline License Agreement

Java SE and NetBeans Cobundle (JDK 8u131 and NB 8.2)		
Product / File Description	File Size	Download
Linux x86	291.05 MB	jdk-8u131-nb-8_2-linux-i586.sh
Linux x64	286.33 MB	jdk-8u131-nb-8_2-linux-x64.sh
Mac OS X x64	342.15 MB	jdk-8u131-nb-8_2-macosx-x64.dmg
Windows x86	319.21 MB	jdk-8u131-nb-8_2-windows-i586.exe
Windows x64	329.42 MB	jdk-8u131-nb-8_2-windows-x64.exe

- License
 - Java SE 8 Readme
 - NB 8.2 3rd Party Readme
 - Installation Instructions

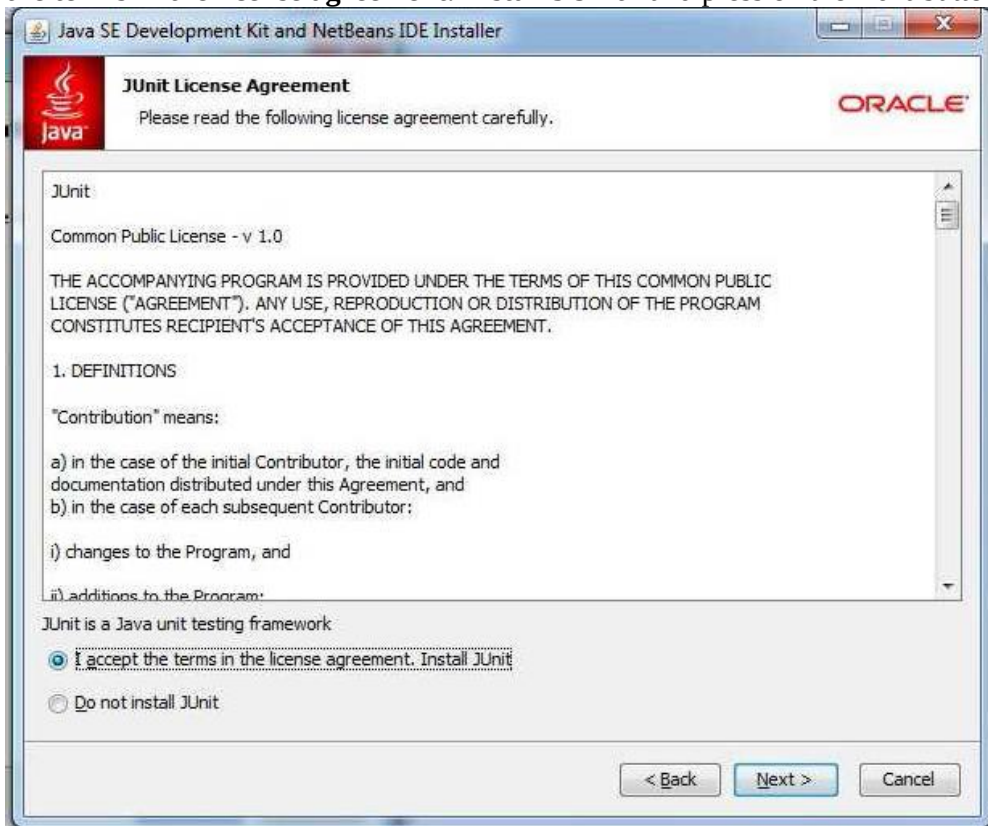
Installation Part 2:

Steps to Install Java and NetBeans

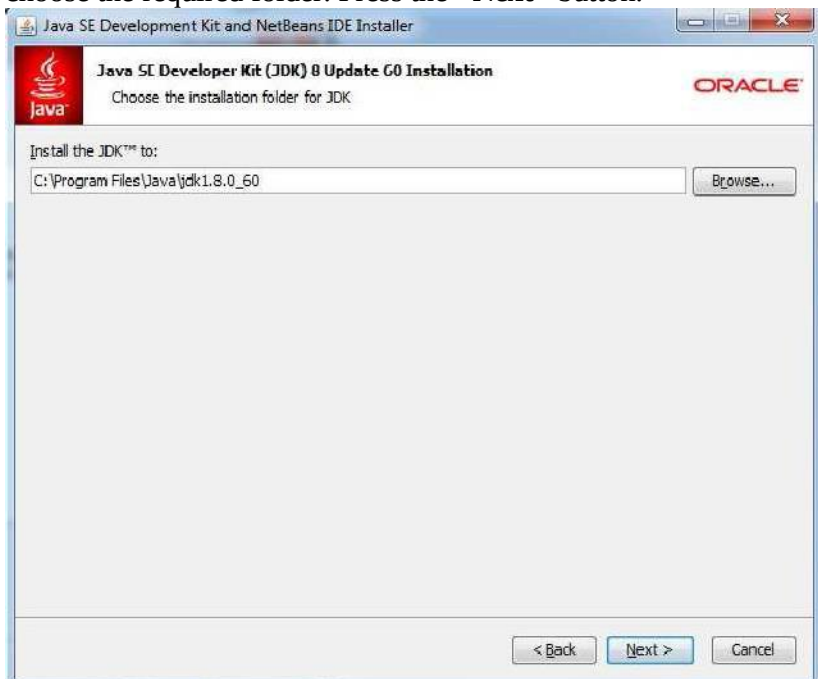
1. Using windows explorer, go to the folder where your NetBeans installer is located.



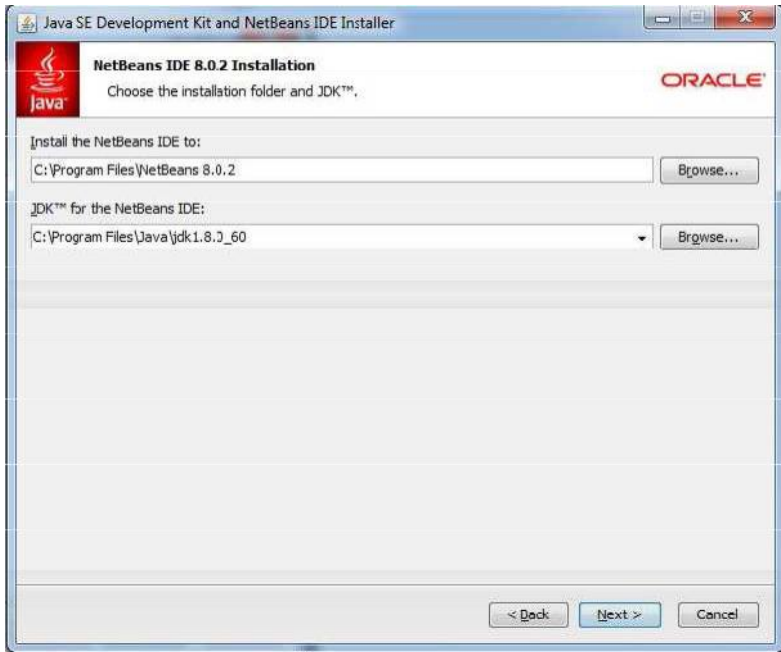
2. Double Click on the installer file, the NetBeans installation wizard window will appear. check the **"I accept the terms in the license agreement. Install JUnit"** and press on the **Next** button



3. From the new window, you can change the installation folder for the JDK by click the *browse* button and choose the required folder. Press the **"Next"** button.



4. The new window allows you to change the installation folder for the NetBeans and JDK by click the corresponding *browse* buttons, then click *Next* button.



5. From the new window, click the **"Install"** button.



6. The installation will start and once it finish the next window will appear. Click the *Finish* button.

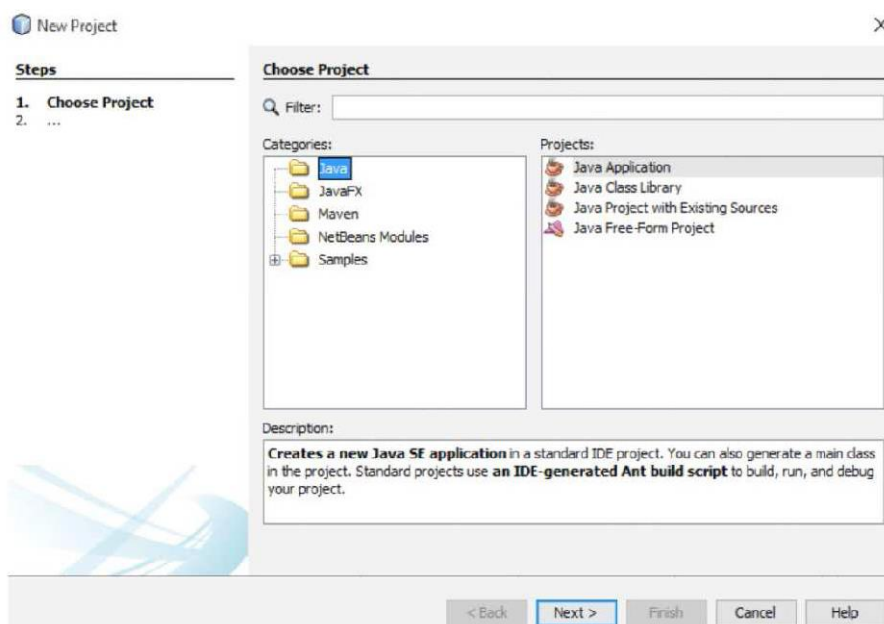


Setting Up the Project

This section provides a very simple and quick introduction to the NetBeans IDE workflow by walking you through the creation of a simple "Welcome to Java!" Java console application. Once you are done with this tutorial, you will have a general knowledge of how to create, build, and run applications in the IDE.

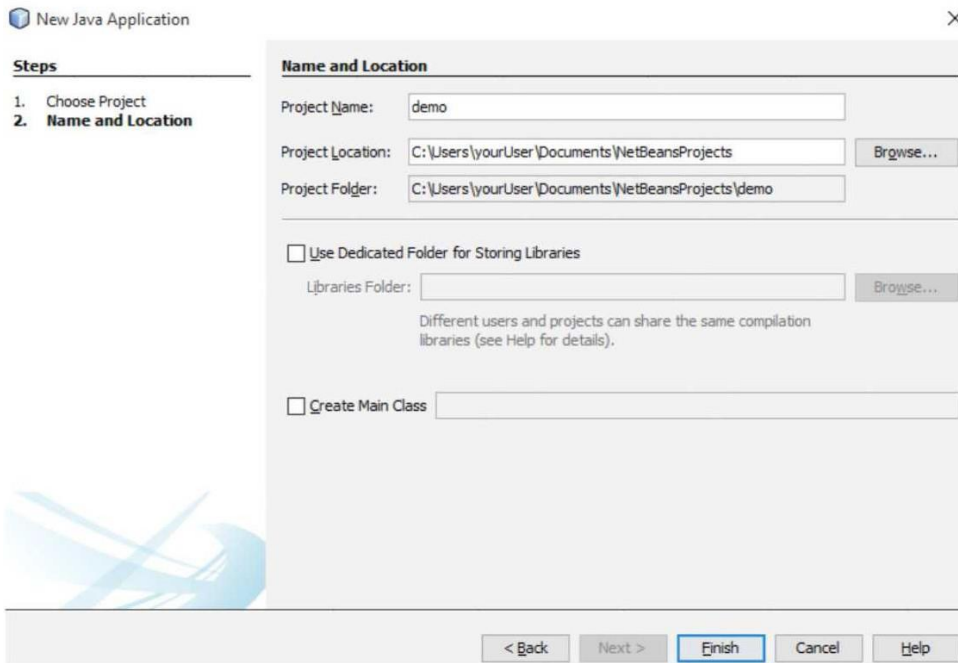
To create an IDE project:

1. Start NetBeans IDE (from the desktop or from the startup menu)
2. Choose File, New Project to display the New Project dialog box, as shown below



3. Select Java in the Categories section and Java Application in the Projects section and then click *Next*, which will display the New Java Application dialog box.

4. We must now name the project. Type **demo** in the Project Name field. Keep the default settings for Project Location and Folder. And uncheck the two check-boxes.

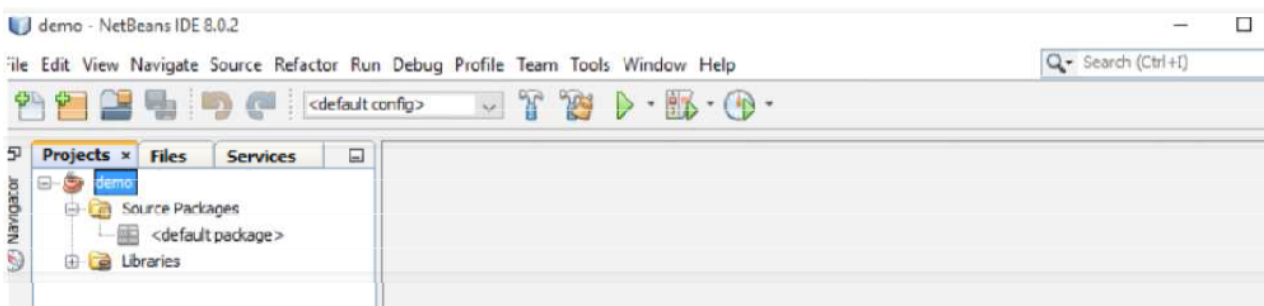


The image shows the 'New Java Application' dialog box in NetBeans. The 'Steps' pane on the left indicates the current step is '2. Name and Location'. The 'Name and Location' section contains the following fields and options:

- Project Name:** demo
- Project Location:** C:\Users\yourUser\Documents\NetBeansProjects (with a 'Browse...' button)
- Project Folder:** C:\Users\yourUser\Documents\NetBeansProjects\demo
- ☐ **Use Dedicated Folder for Storing Libraries**
Libraries Folder: (empty field with a 'Browse...' button)
Different users and projects can share the same compilation libraries (see Help for details).
- ☐ **Create Main Class** (with an empty text field)

At the bottom, there are buttons for '< Back', 'Next >', 'Finish' (highlighted with a blue border), 'Cancel', and 'Help'.

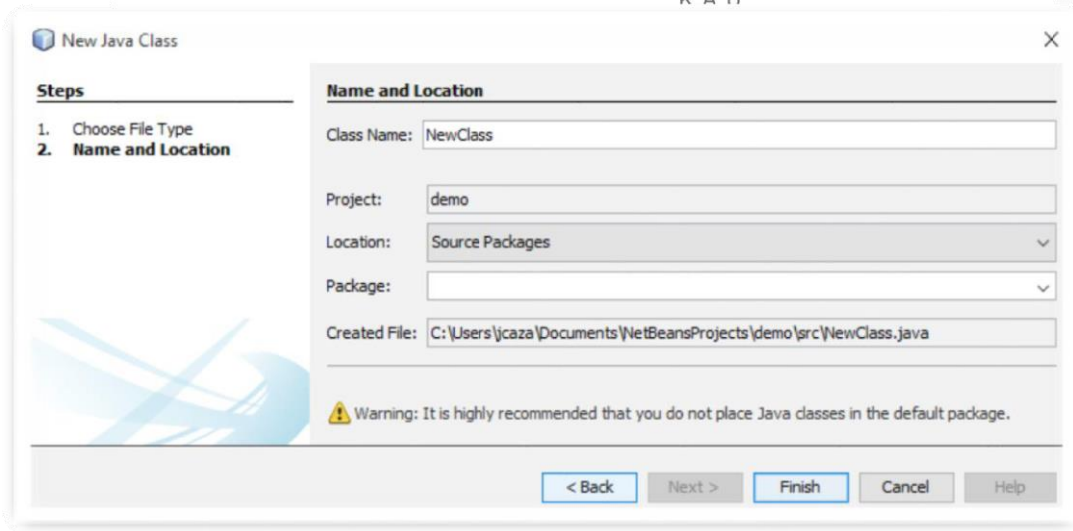
5. Click *Finish* to create the Project, as shown below:



The project is created and opened in the IDE. You should see the **Projects** window, which contains a tree view of the components of the project, including source files, libraries that your code depends on, and so on.

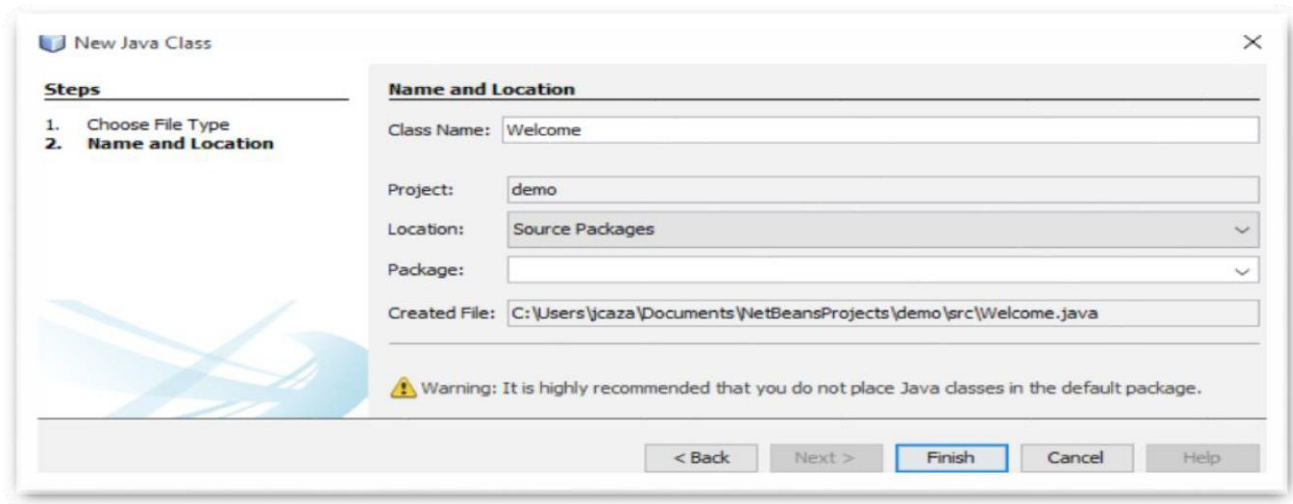
Creating a Java Class in NetBeans

1) From the left side of NetBeans, you can see your new project, **demo**, under the Projects tab. Right-click on the actual project name, “demo”. A menu appears. Choose *New, Java Main Class*, which displays the New Java Class dialog box, as shown below:



The 'New Java Class' dialog box is shown. On the left, under 'Steps', step 2 'Name and Location' is selected. The 'Name and Location' section on the right contains the following fields: 'Class Name' with the value 'NewClass', 'Project' with 'demo', 'Location' with 'Source Packages', an empty 'Package' field, and 'Created File' with the path 'C:\Users\jcaza\Documents\NetBeansProjects\demo\src\NewClass.java'. A warning message at the bottom states: 'Warning: It is highly recommended that you do not place Java classes in the default package.' At the bottom right are buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

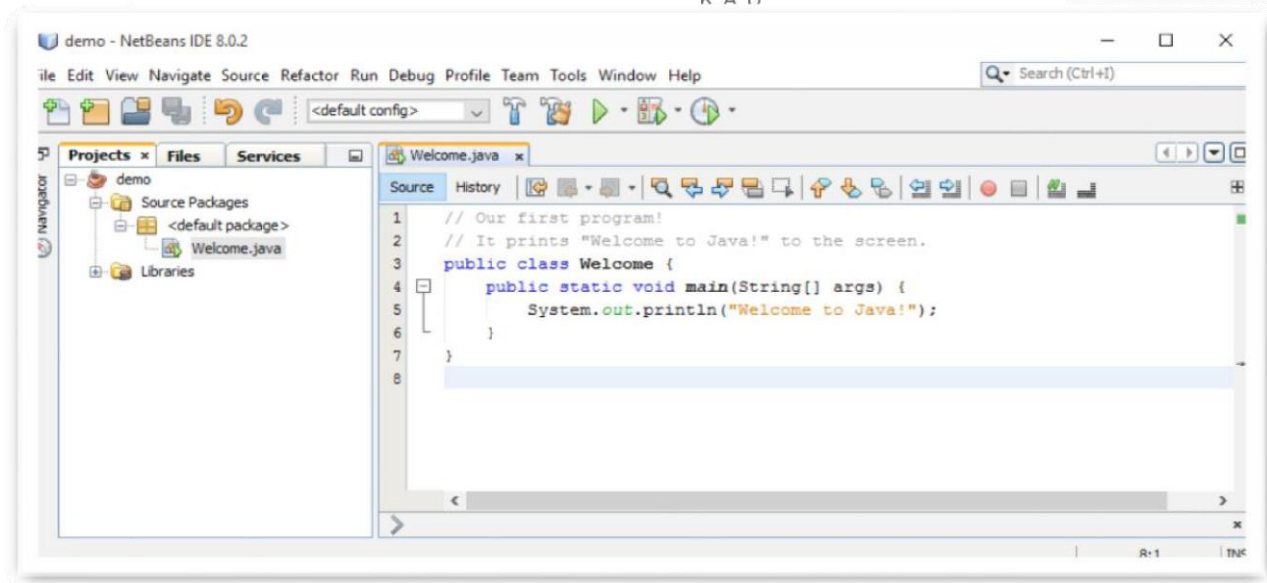
2) You must now give your new Java program a name. Type **Welcome** in the Class Name field. You can leave all other fields as their default values.



The 'New Java Class' dialog box is shown again. In this step, the 'Class Name' field now contains the value 'Welcome'. All other fields ('Project', 'Location', 'Package', 'Created File') remain the same as in the previous screenshot. The warning message and the bottom buttons are also present.

3) Click Finish to create the Welcome class. Because the package field was left empty, the source code for your new program, Welcome.java, will be placed under the <default package> node.

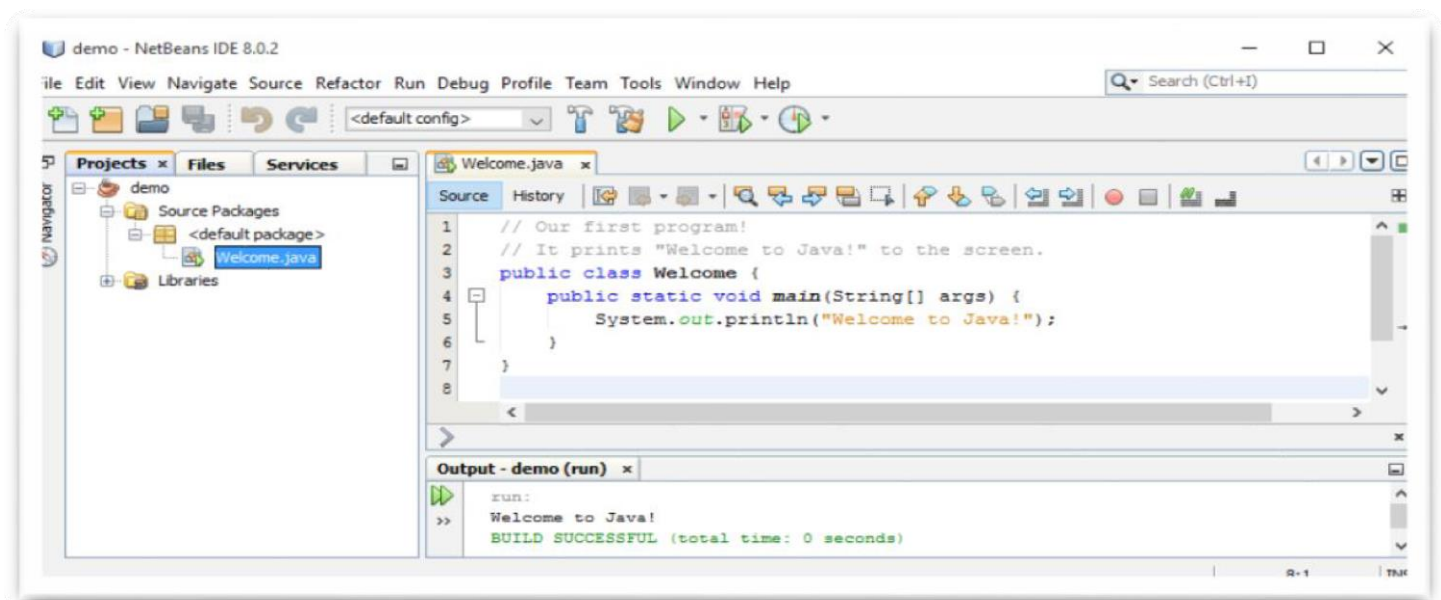
4) Now that we have our first Java program, we can modify the code inside the Welcome class to make it match the figure on the next page.



Running the Program

To run Welcome.java, right-click Welcome.java to display a menu, and then choose **Run File**, or simply press **Shift + F6**.

The Output window opens and displays output similar to what you see in the following figure.



Lab Assessment

Four Problems\ exercises should be completed **in 60 minutes individually**

1. Write a program that displays “**Welcome to Java**” in a single line, but you should use THREE print statements.
(Hint: use print instead of println)

2. Write a program that prints the following pattern:

```
      J      A      V      V      A
      J      A A      V      V      A A
J      J      AAAAA      V V      AAAAA
      J J      A      A      V      A      A
```

3. Write a program that displays the following table:

a	a^2	a^3
1	1	1
2	4	8
3	9	27
4	16	64

4. Write a program that displays the result of

$$\frac{9.5 \times 4.5 - 2.5 \times 3}{45.5 - 3.5}$$